Reference-Free Detection and Suppression of Hallucinations in Low-Resource Neural Machine Translation: A Study on Mizo and Bodo

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by

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Autumn Semester, 2026-27

October 31, 2026

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

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CERTIFICATE

This is to certify that the project report entitled "**Reference-Free Detection and Suppression of Hallucinations in Low-Resource Neural Machine Translation: A Study on Mizo and Bodo**" submitted by **Sharvari Wanjari** (Roll No. 22CS30049) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Master of Technology (Dual Degree) in Computer Science and Engineering is a record of bona fide work carried out by her under my supervision and guidance during Autumn Semester, 2026-27.

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Abstract

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Department: **Department of Computer Science and Engineering**
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Neural machine translation (NMT) systems for low-resource languages sometimes produce *hallucinations*: fluent English that is partly or wholly unrelated to the source. With no reference translation at deployment time, such outputs must be caught from signals inside the translation system. This project studies reference-free hallucination detection and suppression for Mizo (1,525 labelled sentence pairs, 124 hallucinations) and romanised Bodo (1,989 pairs, 119 hallucinations), translated into English with fine-tuned NLLB-200 models. We extract 19 model-internal features (token log-probabilities, cross-attention statistics, logit-lens entropy, hidden-state drift, repetition, length), train a logistic-regression detector and activation probes over the final decoder layer (MAM), and compare them with BLEU, chrF2 and COMET. On 500 held-out sentences per language, the best reference-free detectors reach AUROC 0.896 on Mizo and 0.818 on Bodo, against about 0.95 for COMET, which needs a reference. A detector-gated *repair-or-abstain* policy did not improve translations on average (chrF2 change -0.45 to $+0.02$), although better candidates existed (oracle gain 0.5–2.1); its value lies in abstaining on unreliable outputs, which raises the quality of what is shown by about 7 chrF2 points at 70% coverage. Finally, round-trip features from an independent English→Bodo model raise Bodo AUROC from 0.823 to 0.865.

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Lastly, I would like to acknowledge my own efforts, dedication, and resilience throughout this journey. Completing this thesis has been both a challenging and fulfilling experience.

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Sharvari Wanjari

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Abbreviations

AUROC	Area under the receiver operating characteristic curve
AUPRC	Area under the precision–recall curve
BLEU	Bilingual Evaluation Understudy
chrF2	Character n-gram F-score with $\beta = 2$
COMET	Crosslingual Optimized Metric for Evaluation of Translation
FFN	Feed-forward network (sub-layer of a Transformer block)
LR	Logistic regression
MAM	Model-artifact method: probe over decoder activations
MLP	Multilayer perceptron
NLLB	No Language Left Behind (Meta's NLLB-200 translation models)
NMT	Neural machine translation
PMI	Pointwise mutual information
RT	Round-trip (reverse-model) features
SA / CA	Self-attention / cross-attention

Chapter 1

Introduction

Neural machine translation (NMT) has made it practical to translate between English and languages that, until recently, had almost no machine translation support. Multilingual models such as NLLB-200 (NLLB Team et al., 2022) can be fine-tuned with a few thousand parallel sentences, which brings translation within reach for many low-resource languages of North-East India. Mizo, spoken mainly in Mizoram, and Bodo, a scheduled language of Assam, are two such languages: they have active speaker communities and a growing written presence, but little parallel data and few evaluation resources.

Low-resource NMT systems fail in a characteristic way. Alongside ordinary errors (a wrong word sense, a dropped modifier), they sometimes produce *hallucinations*: fluent, grammatical English that is partly or completely unrelated to the source (Lee et al., 2019; Raunak et al., 2021; Guerreiro et al., 2023c). A Bodo sentence about bamboo architecture becomes "Assamese aviation is a viable means of transportation"; a Mizo dictionary entry for *betel nut* becomes "hammock". Hallucinations are rare, but they are the most harmful kind of error because they look correct. A reader who does not know the source language has no way to notice them, and a single fabricated sentence can mislead in a way a clumsy translation does not.

The standard ways of judging translation quality do not help at deployment time. BLEU (Papineni et al., 2002), chrF (Popović, 2015) and COMET (Rei et al., 2022) all compare the output with a human reference translation, and a reference is exactly what a deployed system does not have. Any practical safeguard must therefore be *reference-free*: it must decide whether to trust a translation using only the source sentence, the output, and the translation model itself.

This project asks how far such reference-free signals can go for Mizo→English and Bodo→English translation, and what a system should do once a hallucination is

suspected. We work with manually labelled outputs of two fine-tuned NLLB-200 (600M, distilled) models: 1,525 Mizo and 1,989 Bodo sentence pairs, each labelled as *correct*, *error*, or one of three hallucination types (fully detached, strongly detached, oscillatory; Guerreiro et al., 2023c). From each translation we extract 19 signals from inside the model: how confident the decoder was, how its cross-attention spread over the source, how uncertain its intermediate layers were, how much the output repeats itself, and how long it is relative to the source. We also capture the raw activations of the final decoder layer and train probes on them, following the idea that a model's internal state carries information about the reliability of what it generates (Snyder et al., 2024; Azaria and Mitchell, 2023).

Detection alone does not change what the user sees. We therefore also build and evaluate a *repair-or-abstain* policy: when the detector flags a translation, the system decodes several alternatives, keeps the first one the detector accepts, and otherwise withholds the output as unreliable. Finally, for Bodo, we train an independent English→Bodo model and ask whether a round trip through it (how well the English output "explains" the original Bodo) adds information the forward model cannot provide about itself.

Research Questions

- **RQ1:** How well do reference-free signals from inside an NMT model detect hallucinations in low-resource Mizo and Bodo translation, compared with reference-based metrics?
- **RQ2:** Which hallucination types do these detectors catch and miss, and what kinds of non-hallucinated translations do they wrongly flag?
- **RQ3:** Can a detector-gated repair-or-abstain policy reduce the harm of hallucinations at inference time, without a reference?
- **RQ4:** Does a separately trained reverse (target→source) model add detection signal beyond the forward model's own internals?

Contributions

1. A hallucination-labelled evaluation set for two low-resource Indian languages translated into English, with a five-way label scheme that separates hallucinations from ordinary translation errors (Chapter 3).

2. A controlled comparison of 22 single-feature detectors, a logistic-regression detector, and eight activation-probe (MAM) variants on identical splits, with thresholds chosen only on validation data (Chapters 4 and 5).

3. An analysis by hallucination type and by false-alarm source, which shows that the detectors mainly separate *bad* translations from good ones rather than hallucinations from ordinary errors (Section 5.4).

4. An end-to-end, reference-free repair-or-abstain pipeline, evaluated against forced-replacement, random and oracle baselines on the same candidate pool (Chapter 6).

5. A round-trip feature set from an independently trained reverse model, with a leakage guard and paired bootstrap estimates of its added value (Chapter 7).

Organisation of the Report

Chapter 2 reviews prior work on hallucination in NMT, reference-free detection, and mitigation. Chapter 3 describes the languages, translation models, annotation scheme and data splits. Chapter 4 defines the features, detectors, threshold selection, suppression policy and round-trip features. Chapters 5, 6 and 7 report the results for detection, suppression and round-trip features respectively. Chapter 8 summarises the findings, discusses limitations and outlines future work. Appendix A gives formal feature definitions and the full experimental configuration.

Chapter 2

Existing Literature and Related Work

2.1 Hallucinations in Neural Machine Translation

Hallucination in natural language generation broadly refers to output that is not supported by the input (Ji et al., 2023). In machine translation the input is fully specified, so the notion is sharper: a hallucination is a translation that is, wholly or partly, *detached* from the source sentence. Lee et al. (2019) showed that NMT models can be pushed into producing fluent but unrelated output by small perturbations of the source, and Raunak et al. (2021) linked natural hallucinations to properties of the training data, such as noisy or misaligned sentence pairs and memorised training examples.

Guerreiro et al. (2023c) carried out the first large-scale study of *natural* hallucinations (those that occur on unperturbed input) and proposed the taxonomy used in this project. *Fully detached* hallucinations have no relation to the source; *strongly detached* hallucinations keep some source content but replace most of it with unsupported material; *oscillatory* hallucinations are degenerate outputs that repeat the same n-gram many times. They also found that natural hallucinations are rare, which makes detection a needle-in-a-haystack problem: methods must be judged on how well they rank the few true cases above a large number of correct and merely erroneous translations. Guerreiro et al. (2023a) extended the analysis to large multilingual models such as NLLB and found that hallucinations become much more frequent for low-resource language pairs, which is the setting of this report.

2.2 Reference-Free Hallucination Detection

2.2.1 Model-Internal Signals

The simplest internal signal is the model's own confidence. *Seq-Logprob*, the length-normalised log-probability of the output under the model that produced it, was found by Guerreiro et al. (2023c) to be a strong baseline that outperformed several previously proposed detectors. A second family uses *attention*: if the decoder is generating content not grounded in the source, its cross-attention should look unusual, for example concentrated on a few tokens such as the end-of-sentence marker. Guerreiro et al. (2023b) formalised this by measuring the optimal-transport distance between the cross-attention distribution and a uniform one (Wass-to-Unif). Dale et al. (2023a) used ALTI+, a method that estimates how much each output token relies on the source versus on the target prefix, and showed that low source contribution is a good hallucination signal. Xu et al. (2023) also introspected source contributions and trained a lightweight classifier on them.

Features of this kind are cheap, need no extra models, and are always available at inference time. Our 19 internal features (Section 4.3) draw on all three ideas: output probability, cross-attention statistics, and the stability of the model's internal representations across layers and positions. The last group is inspired by the *logit lens* (nostalgebraist, 2020), which projects intermediate hidden states through the output layer to see how a prediction forms.

2.2.2 Probing Model Activations

A different line of work trains classifiers directly on hidden activations. Azaria and Mitchell (2023) showed that a small feed-forward classifier on a language model's hidden states can tell whether a statement the model processes is true. Snyder et al. (2024) probed several "artifacts" of a language model—softmax outputs, integrated-gradient attributions, and self-attention and fully connected activations—and found that classifiers on these artifacts can predict hallucinations in factual question answering, in some cases from the very first generated token. Our MAM probes (Section 4.4.3) adapt this idea to an encoder–decoder translation model: we capture the outputs of the self-attention, cross-attention and feed-forward sub-layers of the final decoder layer and train small MLPs on them.

2.2.3 Reference-Based and Similarity-Based Metrics

When a reference translation exists, BLEU (Papineni et al., 2002), chrF (Popović, 2015) and neural metrics such as COMET (Rei et al., 2020, 2022) measure how far the output is from it. These metrics are not available at deployment time, but they are useful in research as an upper bound: they show how well hallucinations can be separated when the right answer is known. Dale et al. (2023a) also showed that cross-lingual sentence similarity between the source and the output (for example with LaBSE) detects hallucinations very well; such external encoders, however, need good coverage of the source language, which is not available for romanised Bodo.

2.2.4 Round-Trip and Reconstruction Signals

A translation that preserves the meaning of the source should allow the source to be recovered. Tu et al. (2017) used reconstruction of the source from decoder states as a training objective, and Moon et al. (2020) revisited round-trip translation (translating the output back and comparing it with the source) as a quality estimate. Zhou et al. (2021) trained token-level hallucination detectors using synthetic data. Chapter 7 applies the round-trip idea with a separately trained reverse model, using both its forced-decoding probability of the original source and its free back-translation.

2.3 Mitigating Hallucinations

Once a hallucination is suspected, a system can either try to produce a better translation or decline to answer. Guerreiro et al. (2023c) and Dale et al. (2023a) showed that generating several candidates (for example with beam search, sampling or Monte Carlo dropout) and reranking them with a detector or a quality-estimation model can remove many hallucinations. Declining to answer is the subject of *selective prediction*: a model abstains on inputs where it is unreliable and is evaluated by its *risk–coverage* trade-off, the error rate on the answered inputs as a function of the fraction answered (Geifman and El-Yaniv, 2017). Our repair-or-abstain policy (Section 4.7) combines both: it repairs when an alternative candidate passes the detector and abstains otherwise, and we report its risk–coverage curve.

2.4 Machine Translation for Mizo and Bodo

NLLB-200 (NLLB Team et al., 2022) covers Mizo (`lus_Latn`) but not Bodo. IndicTrans2 (Gala et al., 2023) covers all 22 scheduled Indian languages, including Bodo in Devanagari script. Benchmarks for hallucination detection exist for a handful of language pairs, including some low-resource ones (Dale et al., 2023b), but, to our knowledge, none include Mizo or Bodo. The data in this project use a romanised (Latin-script) form of Bodo, which neither system supports natively.

2.5 Critical Gaps

- **Language coverage.** Most hallucination-detection studies evaluate on high-resource pairs, or on artificially perturbed inputs. Natural hallucinations in fine-tuned low-resource systems, where they are far more frequent, are less studied.
- **Hallucination versus error.** Many evaluations report a single detection score. In low-resource settings ordinary errors are very common (64% of Bodo outputs in our data), so it matters whether a detector finds hallucinations specifically or simply flags any poor translation.
- **From detection to action.** Detection is often evaluated in isolation. Whether a reference-free detector can drive a repair or abstention decision that actually improves the output, compared with simple baselines, is rarely measured end to end.
- **Protocol.** Thresholds are often tuned on the test set, and small test sets are reported without confidence intervals. With 54–56 hallucinations in each test set, both issues can change conclusions.

This project addresses these gaps with a fixed train/validation/test protocol, subtype-level and false-alarm analysis, an end-to-end suppression evaluation with baselines, and bootstrap confidence intervals.

Chapter 3

Data and Annotation

3.1 Languages and Translation Systems

Mizo (also called Lushai; ISO 639-3 `lus`) is a Kuki-Chin language written in the Latin script with a few diacritics (for example *â*, *ṭ*). It is covered by NLLB-200 under the code `lus_Latn`. The Mizo→English system in this project is NLLB-200-distilled-600M fine-tuned on Mizo–English parallel data (checkpoint `nllb-mizo-eng-full`).

Bodo (`brx`) is a Tibeto-Burman language of Assam and one of the 22 scheduled languages of India. It is normally written in Devanagari. The data used here are *romanised*: a Latin-script transliteration of Devanagari that keeps distinctions such as vowel length through capitalisation (e.g. *phAlAMgi*, *khAlAmanAyani*) and keeps the danda (|) as the sentence terminator. Bodo is not an NLLB-200 language, so the Bodo→English system (`nllb-bodo-roman-to-eng`) was obtained by fine-tuning NLLB-200-distilled-600M on a Bodo–English training split of 10,244 sentence pairs, reusing the Mizo language token (`lus_Latn`) as a stand-in source-language code. Because the model is fine-tuned, the choice of code only needs to be consistent between training and inference.

Both models translate into English (`eng_Latn`). The outputs analysed in this report were produced once by these systems and then labelled; all features are computed by re-running the same checkpoint in teacher-forced mode on the stored (source, output) pairs (Section 4.7.4).

3.2 Annotation Scheme

Each translation was labelled by comparing the source, the English reference, and the model output. The scheme separates hallucinations from ordinary translation errors, following the taxonomy of Guerreiro et al. (2023c):

correct The output conveys the meaning of the source; small stylistic differences from the reference are allowed.

error The output is related to the source but contains mistranslations, omissions or additions that do not make it detached as a whole.

hallucination, fully detached The output has essentially no relation to the source.

hallucination, strongly detached Some source content survives (a name, a topic, a sentence frame), but most of the output is unsupported by the source.

hallucination, oscillatory The output degenerates into repetition of the same word or phrase.

For detection, the three hallucination labels form the positive class and *correct* and *error* form the negative class. Keeping *error* in the negative class is deliberate: a useful detector must flag fabricated content without also flagging every imperfect translation. Mizo rows labelled *unclear* (9) or *hallucination (subtype disputed)* (8) were removed rather than forced into a class, leaving 1,525 Mizo pairs. For Bodo, 1,989 of the 2,000 labelled pairs were retained after cleaning.

Table 3.1 shows one example of each hallucination type.

3.3 Dataset Statistics

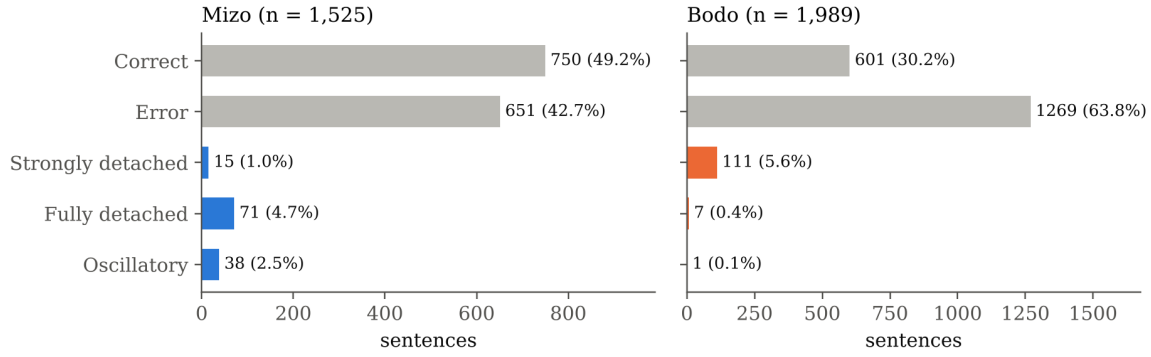
Figure 3.1 shows the label distribution. The two languages differ in ways that matter for detection:

- **Hallucination mix.** Mizo hallucinations are mostly *fully detached* (71) and *oscillatory* (38), with only 15 strongly detached. Bodo hallucinations are almost all *strongly detached* (111 of 119): the output keeps a topic or frame from the source and fills the rest with fabricated content. Bodo has a single oscillatory example.
- **Error rate.** 64% of Bodo outputs are labelled *error*, against 43% for Mizo. Bodo detectors must therefore separate hallucinations from a much larger pool of imperfect but non-hallucinated translations.

TABLE 3.1: Examples of the three hallucination types from the labelled data.

Fully detached (Bodo, R0284)	
Source	*mosauni donakhomAnAyakhau gobAM samAvano sAmabrAni dihunathAyaphora bAnAyanAyAva bAhAyanAya jAyo|*
Reference	The cow's hide is often used to make leather products.
Output	Coconut oil is often used in the manufacture of coconut products.
Strongly detached (Bodo, R0286)	
Source	*asa'miyA auvAni buminasArA dAthAya lunAyani monase arAyathA rAhA|*
Reference	Assamese bamboo architecture is a sustainable way of building structures.
Output	Assamese aviation is a viable means of transportation.
Oscillatory (Mizo, R3393)	
Source	*A chung zâwl te, a sîr te, a kîte chu rangkachak hlîr hlakin luan tûr a ni, a tlâng chu rangkachaka luan tûr a ni.*
Reference	You shall overlay it with pure gold, its top, its sides around it, and its horns; and you shall make a gold molding around it.
Output	You shall overlay it with pure gold on the top, and on the top, and on the top, and on the top, and on the top of it.

- **Domain and length.** The Mizo data mix Bible verses, religious publications, conversational phrases, literary lines and dictionary-style lexical entries (6.1% of Mizo sources have three words or fewer). Fully detached Mizo hallucinations are concentrated in these short entries: their sources average only 6.4 words. The Bodo data are mainly news headlines and general-knowledge sentences of moderate length.



Grey: non-hallucination labels. Coloured: the three hallucination subtypes (positive class).

FIGURE 3.1: Label distribution of the two labelled datasets.

Table 3.2 gives average lengths and reference-based scores per label. Oscillatory outputs are about twice as long as their references, as expected from repetition. Hallucinations of every type have much lower chrF2 and COMET than correct translations, but strongly detached hallucinations sit between the extremes: on Bodo their mean COMET is 0.55,

against 0.72 for errors and 0.43 for fully detached outputs, so many of them score in the same range as ordinary errors.

TABLE 3.2: Mean source, reference and output length (words) and mean reference-based scores per label.

Language	Label	n	Src	Ref	Out	BLEU	chrF2	COMET
Mizo	Correct	750	18.6	17.5	16.5	36.4	56.7	0.75
	Error	651	21.1	20.2	19.0	19.2	40.3	0.62
	Fully detached	71	6.4	6.2	6.6	7.8	12.2	0.44
	Strongly detached	15	20.6	18.3	19.7	8.1	27.4	0.49
	Oscillatory	38	21.9	21.5	42.6	7.0	24.4	0.41
Bodo	Correct	601	12.5	13.9	13.9	55.3	76.8	0.85
	Error	1269	14.3	15.7	15.7	31.6	57.1	0.72
	Fully detached	7	6.0	7.7	9.4	8.4	19.3	0.43
	Strongly detached	111	14.3	16.3	16.3	13.6	36.3	0.55
	Oscillatory	1	28.0	15.0	55.0	0.7	10.1	0.26

3.4 Data Splits

Each language is split once, with a fixed random seed (42), into three disjoint parts (Table 3.3):

- **Test** (500 sentences): 45% of all hallucinations are placed here and the rest is filled with randomly drawn non-hallucinations. Over-representing hallucinations in the test set gives enough positive examples (56 Mizo, 54 Bodo) for meaningful precision and recall estimates.
- **Validation**: 20% of the remaining rows, containing 20% of the remaining hallucinations. Validation data are used only to choose decision thresholds.
- **Fit**: everything else. Fit data are used to train detectors and to learn the direction of each single feature.

The test set is scored once per detector. Because the test set is enriched with hallucinations (11% versus 6–8% in the full data), precision values on it are higher than they would be at the natural hallucination rate; AUROC is unaffected by this.

The validation sets contain only 14 (Mizo) and 13 (Bodo) hallucinations. Thresholds chosen on so few positives are noisy, and Section 5.3 shows cases where this matters. The round-trip experiments in Chapter 7 use a separate, label-stratified split of the Bodo data with the same sizes (500 test rows, 54 test hallucinations).

TABLE 3.3: Fit / validation / test splits (seed 42). H = hallucinations.

	Mizo (1,525 rows, 124 H)			Bodo (1,989 rows, 119 H)		
Split	Rows	H	H rate	Rows	H	H rate
Fit	820	54	6.6%	1,191	52	4.4%
Validation	205	14	6.8%	298	13	4.4%
Test	500	56	11.2%	500	54	10.8%

Test-set composition

Mizo: 243 correct, 201 error, 33 fully detached, 7 strongly detached, 16 oscillatory

Bodo: 142 correct, 304 error, 2 fully detached, 52 strongly detached, 0 oscillatory

Chapter 4

Methodology

4.1 Overview

Figure 4.1 summarises the system. A source sentence is translated by the fine-tuned NLLB model. The (source, output) pair is then passed once more through the same model in teacher-forced mode, which yields the model's token probabilities, attention maps and hidden states for exactly the output it produced. From these we compute 19 scalar features and capture final-layer decoder activations. A detector turns these signals into a hallucination score, and a threshold chosen on validation data turns the score into a decision. Unflagged outputs are returned unchanged. Flagged outputs enter the suppression stage, which either repairs the translation with an alternative candidate that passes the detector, or abstains and marks the output as unreliable. Reference-based metrics (BLEU, chrF2, COMET) appear only on the evaluation side: they are used to compare detectors and to judge repairs, never as detector inputs at inference time.

4.2 Reference-Based Metrics

Three reference-based metrics serve as comparison points and as evaluation tools: sentence-level BLEU (Papineni et al., 2002) with effective n-gram order, chrF2 (Popović, 2015) (character n-gram F-score with $\beta = 2$), both computed with sacreBLEU (Post, 2018), and COMET (Rei et al., 2022) with the `Unbabel/wmt22-comet-da` model, which scores the (source, output, reference) triple. Each is turned into a hallucination score by negating it (lower similarity $\Rightarrow$ more suspicious). COMET was run in a separate environment because its pinned dependencies conflict with the version of `transformers` used for feature extraction.

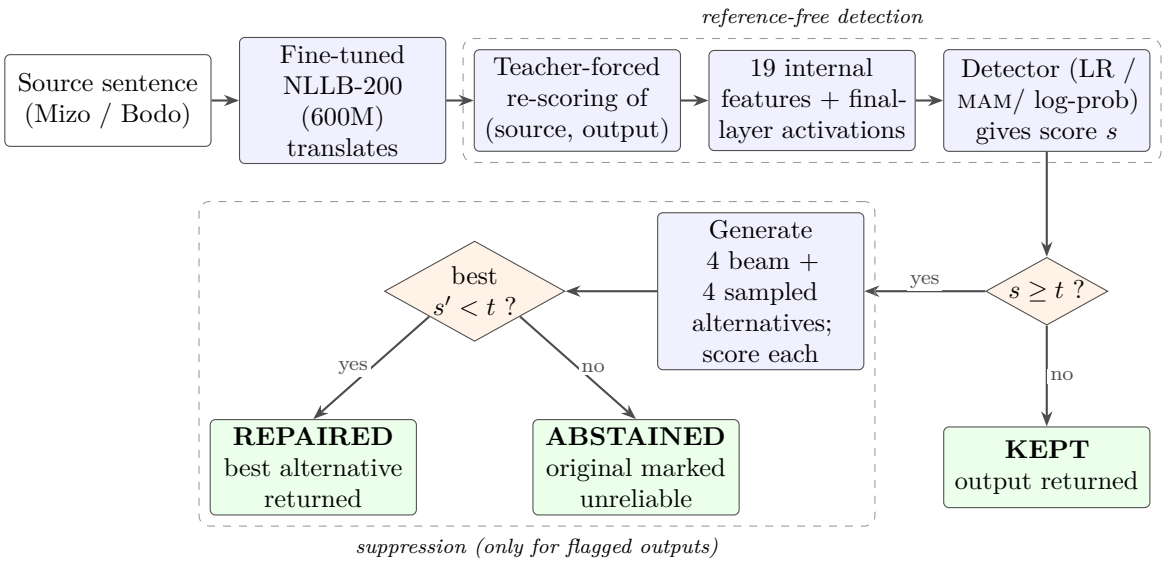


FIGURE 4.1: The reference-free detect–repair–abstain pipeline. The threshold t is chosen on validation data. References are used only for offline evaluation.

4.3 Internal Features

All internal features are computed in a single teacher-forced forward pass. The source is encoded, the stored output (prefixed with the English language token) is given as the decoder target, and we request the output logits, all cross-attention maps and all decoder hidden states. The forced language token at position 0 is excluded from every statistic. Inputs are truncated to 128 tokens. Table 4.1 lists the features by group; Appendix A gives the formulas.

4.4 Detectors

All detectors output a score where *higher means more likely to be a hallucination.*

4.4.1 Single-Feature Detectors

Each of the 19 internal and 3 reference-based features is used on its own. Its direction (whether high or low values indicate hallucination) is learned on the fit split: the feature is negated if its fit-split AUROC is below 0.5. Missing values (20 Mizo rows have an undefined `staticity_w3` because the output is too short) are filled with the fit-split median.

TABLE 4.1: The 19 model-internal features, grouped by the signal they capture.

Feature	Definition (short)
Output confidence	
`seq_logprob`	Mean log-probability of the output tokens (length-normalised Seq-Logprob).
`min_token_logprob`	Log-probability of the least likely output token.
Cross-attention to the source (final decoder layer, averaged over heads)	
`source_entropy`	Mean normalised entropy of each output token's attention over source tokens.
`source_concentration`	Mean of the largest attention weight per output token.
`top20_mass`	Mean attention mass on the top 20% of source tokens.
`high_contrib_ratio`	Share of source tokens that receive total attention ≥ 0.5.
`staticity_w1/w2/w3`	Mean cosine similarity between attention rows $w = 1, 2, 3$ steps apart (attention that stops moving).
`staticity_max`	Maximum of the three staticity values.
Internal representations	
`anchor_layer_entropy_mean`	Mean entropy of the next-token distribution obtained by projecting decoder layers 6, 9 and 12 through the output layer (logit lens).
`anchor_layer_entropy_min`	Minimum of the same three entropies.
`subword_drift_mean`	Mean cosine distance between consecutive final-layer hidden states.
Surface form	
`src_token_len`, `hyp_token_len`	Source and output length in subword tokens.
`hyp_length`	Output length in words.
`length_ratio`	Output tokens / source tokens.
`ngram_repetition_2/3`	1− (unique / total) word bigrams and trigrams in the output.

4.4.2 Logistic Regression over Internal Features

The logistic-regression (LR) detector combines all 19 internal features. Features are median-imputed and standardised with statistics from the fit split. The model uses L_2 regularisation with $C = 1$ and balanced class weights to counter the 5–7% positive rate in the fit split. Its predicted probability is the hallucination score. It uses no reference-based feature, so it can run at inference time.

4.4.3 Activation Probes (mam)

The MAM (model-artifact method) probes follow the idea of classifying hallucinations from a model's internal activations (Snyder et al., 2024; Azaria and Mitchell, 2023).

Forward hooks on the *final* decoder layer capture the outputs of its three sub-layers:

- self-attention (SA, `self_attn`),
- cross-attention to the encoder (CA, `encoder_attn`),
- the feed-forward block (FFN, `fc2`),

each a 1,024-dimensional vector per decoder position. We read them at two positions: **first**, the state that predicts the first output token, and **last**, the state at the end-of-sentence token, which has seen the whole output. This gives eight probe variants: SA, CA, FFN and their concatenation (*combined*, 3,072 dimensions), each at the first or last position.

Each probe is a one-hidden-layer MLP (256 ReLU units, one logit output). Inputs are standardised with the fit-split mean and standard deviation. The probe is trained on the fit split for 1,000 iterations with mini-batches of 128, Adam (learning rate 10^{-4}, weight decay 10^{-2}), and binary cross-entropy with the positive class weighted by the negative-to-positive ratio. The sigmoid output is the hallucination score.

4.5 Choosing Decision Thresholds

A score becomes a decision with a threshold t: flag the output if $s \geq t$. For every detector we choose t on the *validation* split with three rules and then apply it unchanged to the test split:

- **Youden**: maximise TPR − FPR (Youden, 1950), a balanced operating point.
- **F1**: maximise the F1 score of the hallucination class.
- **Recall**: among thresholds with recall ≥ 0.8, take the one with the highest precision (ties broken towards higher recall).

4.6 Evaluation Metrics for Detection

Threshold-free. AUROC measures how well a detector ranks hallucinations above non-hallucinations; AUPRC (average precision) emphasises the top of the ranking and is more sensitive to the rarity of the positive class. Its value for a random detector equals the positive rate (0.11 on our test sets).

At a threshold. Precision, recall, F1 and specificity (the share of non-hallucinations left unflagged) on the test split.

By subtype and by negative class. Recall is computed separately for each hallucination type, with 95% Wilson score intervals (Wilson, 1927) because some types have very few test examples. AUROC is also computed for each type against all non-hallucinations. The false-alarm rate is reported separately for *correct* and *error* translations, to see which kind of negative a detector confuses with hallucinations.

4.7 Suppression: Repair or Abstain

4.7.1 Candidate Generation

For each test sentence the same NLLB checkpoint produces alternative translations in two ways, with settings fixed before any evaluation:

- the 4 best hypotheses from beam search with beam width 8;
- 4 samples with nucleus sampling (Holtzman et al., 2020) (top-$p = 0.9$, temperature 0.7).

Duplicates and candidates identical to the original output are removed. Every candidate is scored with both suppression detectors: the log-probability detector (score $= -$`seq_logprob`) and the MAM combined [last] probe. These two were chosen because they are the strongest single-model detectors that can score an arbitrary new candidate in one forward pass.

4.7.2 The Gated Policy

Given a detector D and threshold t, each sentence ends in one of three states:

- **KEPT**: the original output scores $s < t$; it is returned unchanged.
- **REPAIRED**: the original is flagged, and the lowest-scoring alternative has $s' < t$; that alternative is returned.
- **ABSTAINED**: the original is flagged and no alternative passes; the original is returned but marked unreliable (in a product it would be withheld or shown with a warning).

Because unflagged sentences are never changed, applying the gate offline to a pre-generated candidate pool gives the same outcome as generating candidates only for flagged sentences. This lets us evaluate all three threshold rules on one pool.

4.7.3 Baselines and Evaluation

The policy is compared with three alternatives applied to the *same flagged sentences*:

- **forced_best**: always replace with the lowest-scoring alternative, even if it is still flagged (no abstention);
- **random**: replace with a randomly chosen alternative;
- **oracle**: replace with the alternative that is best according to the reference metric. This is an upper bound on what any selector could achieve with this candidate pool.

Repairs are judged against the reference translation, which the detectors never see. The planned primary metric was COMET; in both runs the COMET scoring step failed (for Mizo, the GPU ran out of memory), so the scripts fell back to **chrF2**, which is used throughout Chapter 6. We report the mean change in chrF2 (final − original) over all 500 test sentences and separately over true hallucinations and non-hallucinations, with 95% bootstrap confidence intervals (2,000 resamples). We also plot risk–coverage curves (Geifman and El-Yaniv, 2017): mean chrF2 of the answered (non-abstained) sentences against the fraction answered.

4.7.4 Score Parity Check

A gated policy is only meaningful if the detector applied to new candidates computes exactly the same score as the detector whose threshold was tuned. Before generating candidates, the suppression script therefore re-scores the original outputs and stops unless the recomputed scores match the stored ones. On the final runs, the maximum absolute difference over all 500 test sentences was 0.00000 for both detectors and both languages. An earlier audit of the Mizo pipeline found that the stored internal features did not match the current model checkpoint (log-probability differences up to 0.55), so the Mizo internal features were re-extracted with the current checkpoint and all Mizo detectors were re-trained before the suppression experiments.

The script also reports how the stored outputs were decoded: re-decoding 50 test sources with 4-beam search reproduced 94% of the stored Bodo outputs exactly and 50% of the

Mizo outputs (greedy decoding: 12% and 16%). The Mizo outputs were thus produced with decoding settings that differ from 4-beam search; this does not affect detection, which scores the stored outputs, but it means the "original" in a re-run pipeline is not always identical to the labelled one.

4.8 Round-Trip Features

For Bodo we train a reverse model, English→romanised Bodo, by fine-tuning NLLB-200-distilled-600M from the public base checkpoint (not from the forward model) on the English–Bodo training pairs. A leakage guard removes any training pair whose Bodo side matches a labelled source, or whose English side matches a labelled reference or output, after Unicode and whitespace normalisation; no pairs overlapped, so all 10,244 pairs were used. Training ran for 3 epochs with an effective batch of 32 (8×4 accumulation steps), learning rate 5×10^{-5} with linear decay and 20 warm-up steps, Adafactor, and mixed precision. The NLLB code `asm_Beng` stands in for the Bodo target language.

Given an English output y and its Bodo source x, the reverse model provides seven features:

- `rt_lp_sum`, `rt_lp_mean`, `rt_lp_min`: sum, mean and minimum token log-probability of x given y (forced decoding);
- `rt_pmi_mean`: $\frac{1}{|x|}\left[\log p(x \mid y) - \log p(x \mid \varnothing)\right]$, how much y raises the probability of x compared with an empty input;
- `rt_chrf`, `rt_bleu`: chrF and BLEU between a 4-beam back-translation of y and x;
- `rt_len_ratio`: length of the back-translation over length of x (characters).

These are evaluated as single features and inside logistic regressions (round-trip only; base internal; base + round-trip) on a label-stratified split (500 test, 298 validation, 1,191 train). AUROC confidence intervals use 1,000 bootstrap resamples, and the gain from adding round-trip features is tested with 2,000 paired bootstrap resamples of the test set.

4.9 Implementation

All experiments ran on Kaggle notebooks with NVIDIA T4 GPUs, using PyTorch, Hugging Face `transformers` (Wolf et al., 2020), `scikit-learn` (Pedregosa et al., 2011),

sacreBLEU and `unbabel-comet`. Every stage writes its outputs to disk (feature tables, split assignments, detector scores, thresholds, trained probes and candidate pools), so later stages can be re-run in a fresh session without recomputation. Appendix A.2 lists all settings.

Chapter 5

Results: Hallucination Detection

5.1 Overview

This chapter answers RQ1 and RQ2. All numbers are on the 500-sentence test split of each language (56 Mizo and 54 Bodo hallucinations), with detectors trained on the fit split and thresholds chosen on the validation split (Section 3.4). We first compare all detectors without a threshold (AUROC, AUPRC), then at concrete operating points, then by hallucination type and false-alarm source, and finally look at individual errors.

5.2 Threshold-Free Comparison

5.2.1 Single Features

Figure 5.1 shows the test AUROC of every single feature in both languages, together with the two learned detectors. Three patterns stand out.

- **Reference-based metrics are the strongest signals.** COMET reaches 0.958 (Mizo) and 0.948 (Bodo). This is the ceiling a reference-free system is measured against, not something it can use.
- **Output confidence transfers across languages.** `seq_logprob` is almost equally useful in both (0.799 Mizo, 0.805 Bodo), and `min_token_logprob` is close behind (0.757, 0.730). This matches its role as a strong baseline in high-resource studies (Guerreiro et al., 2023c).
- **Other signals are language-specific.** Logit-lens entropy is the best internal feature for Mizo (0.828) but only fifth for Bodo (0.722). Cross-attention features

behave the other way: `source_entropy` reaches 0.760 on Bodo but 0.586 on Mizo, and `source_concentration` 0.729 against 0.576. Repetition features are informative only for Mizo (`ngram_repetition_3`: 0.615 against 0.496), which has 16 oscillatory test hallucinations while Bodo has none. Source length helps on Mizo (0.660), because fully detached Mizo hallucinations are concentrated in very short dictionary-style entries.

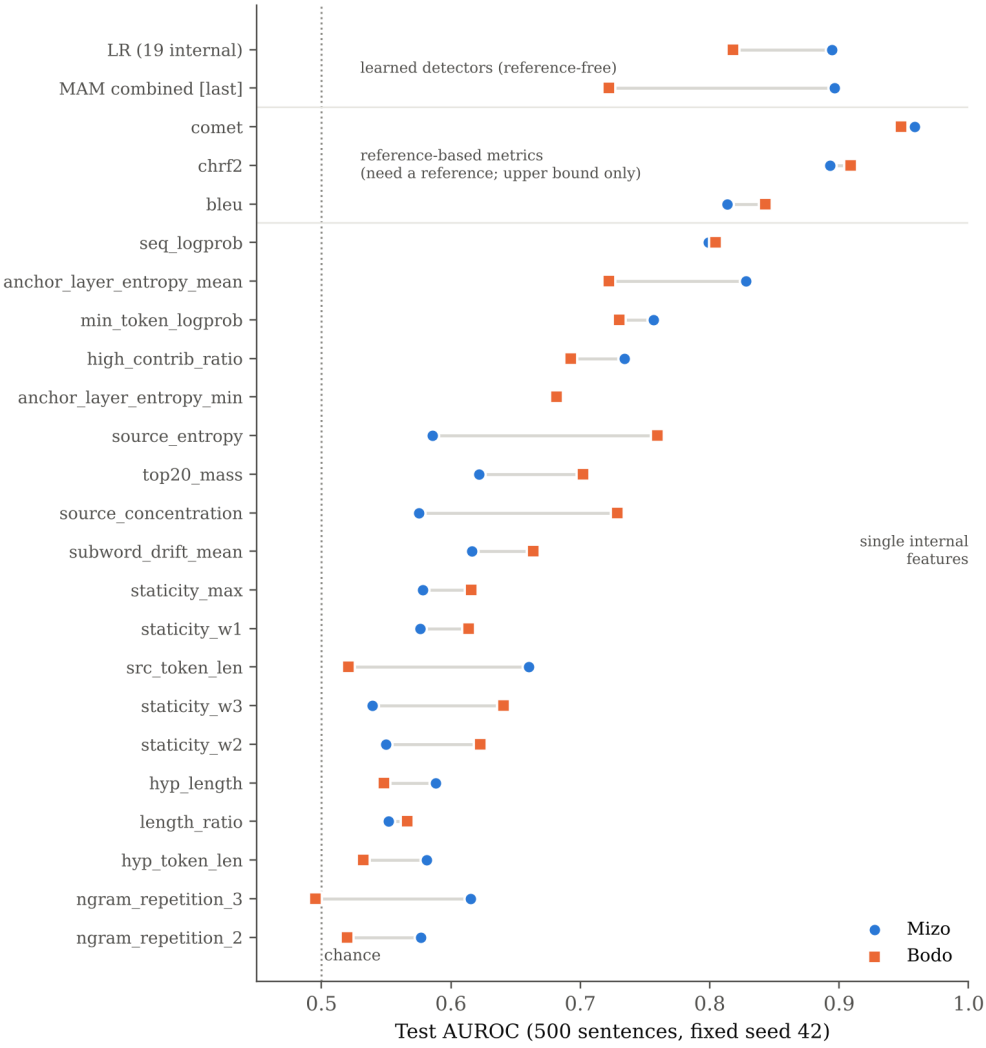


FIGURE 5.1: Test AUROC of every single feature and of the two learned reference-free detectors. Lines connect the Mizo and Bodo values of the same detector.

Figure 5.2 shows why the strongest features work: hallucinated outputs have lower mean log-probability and higher logit-lens entropy in both languages, but the distributions overlap substantially. The length ratio is shifted only slightly. COMET separates the classes much more cleanly.

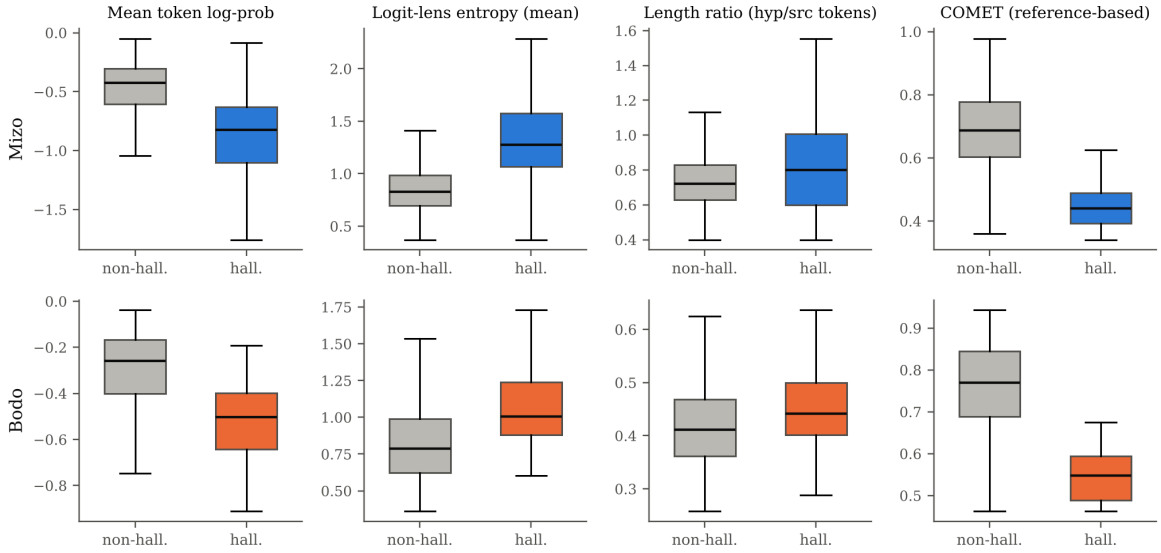


FIGURE 5.2: Distribution of four features for non-hallucinated (grey) and hallucinated outputs (whiskers at 1.5 IQR, values clipped to the 1st–99th percentile). Mizo internal features are shown from the released feature file of the first extraction pass (Section 4.7.4).

5.2.2 Learned Detectors

Table 5.1 compares the learned detectors with the strongest single features. On **Mizo**, the two learned reference-free detectors are effectively tied: the MAM combined [last] probe reaches AUROC 0.896 and LR 0.894, both above the best single internal feature (0.828) and level with chrF2 (0.893), a reference-based metric. LR has the higher AUPRC (0.688 against 0.644), meaning it ranks the most confident flags more precisely. On **Bodo**, LR is the best reference-free detector (0.818), only modestly above `seq_logprob` alone (0.805), and the MAM probes fall well behind (0.722).

5.2.3 Activation Probes

Table 5.2 lists all eight MAM variants. In both languages the **last** decoder position is generally more informative than the first: the best variant is *combined [last]* in both, and on Mizo every last-position variant beats its first-position counterpart. This differs from the early-detection setting of Snyder et al. (2024), where the first generated token already carries signal. In translation, the end-of-sentence state has attended to the whole output, so it can reflect whether the output as a whole stayed grounded; the state before the first token has not yet committed to an output.

The gap between the languages is much larger for the probes (0.896 against 0.722) than for LR (0.894 against 0.818) or log-probability (0.799 against 0.805). Two explanations fit the data. First, the hallucination types differ: Mizo hallucinations are dominated by degenerate, repetitive or default outputs, which leave a distinctive trace in the decoder

TABLE 5.1: Test AUROC and AUPRC of the reference-based metrics, learned detectors and strongest internal features. Best reference-free value per column in bold. Random AUPRC ≈ 0.11.

		Mizo		Bodo	
Family	Detector	AUROC	AUPRC	AUROC	AUPRC
Reference-based	COMET	0.958	0.787	0.948	0.686
	chrF2	0.893	0.547	0.909	0.482
	BLEU	0.814	0.439	0.843	0.402
Learned	LR (19 internal features)	0.894	**0.688**	**0.818**	0.297
	MAM combined [last]	**0.896**	0.644	0.722	**0.330**
	MAM FFN [last]	0.887	0.602	0.712	0.301
Single internal	`seq_logprob`	0.799	0.458	0.805	0.248
	`anchor_layer_entropy_mean`	0.828	0.507	0.722	0.196
	`min_token_logprob`	0.757	0.244	0.730	0.196
	`high_contrib_ratio`	0.734	0.413	0.693	0.211
	`source_entropy`	0.586	0.175	0.760	0.270
	`source_concentration`	0.576	0.310	0.729	0.245

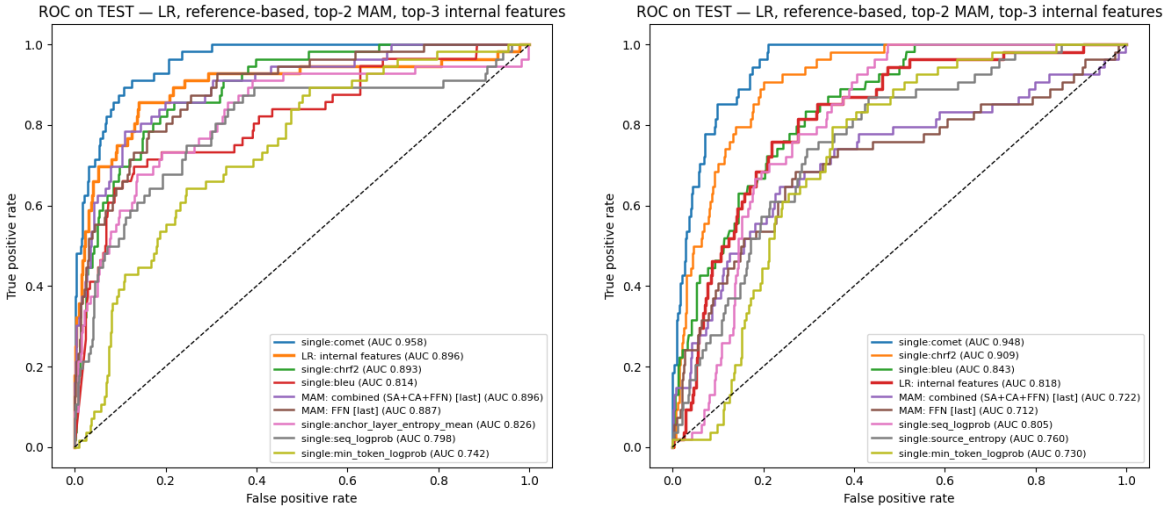
TABLE 5.2: Test AUROC / AUPRC of the eight MAM probe variants (final decoder layer).

	Mizo		Bodo	
Variant	AUROC	AUPRC	AUROC	AUPRC
Self-attention [first]	0.848	0.616	0.684	0.235
Cross-attention [first]	0.858	0.592	0.708	0.282
FFN [first]	0.863	0.616	0.677	0.252
Combined [first]	0.859	0.586	0.679	0.279
Self-attention [last]	0.881	0.587	0.685	0.219
Cross-attention [last]	0.864	0.629	0.705	0.312
FFN [last]	0.887	0.602	0.712	0.301
Combined [last]	**0.896**	**0.644**	**0.722**	**0.330**

state, whereas Bodo hallucinations are almost all strongly detached sentences that are fluent and on-topic. Second, each probe learns 3,072 input dimensions from only 52 positive Bodo examples, against a 19-dimensional LR that is far less prone to overfitting.

5.2.4 ROC Curves

Figure 5.3 shows ROC curves for the main detectors. On Bodo, COMET reaches full recall at a false-positive rate of about 0.2, while the reference-free detectors need a much higher false-positive rate before they approach full recall. On Mizo the reference-free curves rise more steeply and sit close to chrF2.



(A) Mizo (B) Bodo

FIGURE 5.3: Test ROC curves for the reference-based metrics, LR, the two best MAM probes and the three best internal features. The Mizo curves for internal-feature detectors are from the first extraction pass; their AUROCs differ from Table 5.1 by at most 0.015.

5.3 Operating Points

Table 5.3 gives test precision, recall, F1 and specificity for the four main detectors at the three validation-chosen thresholds.

TABLE 5.3: Test metrics at thresholds chosen on validation data. P = precision, R = recall, Sp = specificity.

		Mizo				Bodo			
Detector	Rule	P	R	F1	Sp	P	R	F1	Sp
LR	Youden	0.48	0.73	0.58	0.90	0.29	0.74	**0.42**	0.78
	F1	0.73	0.54	**0.62**	0.98	0.32	0.59	**0.42**	0.85
	Recall	0.36	0.89	0.51	0.80	0.30	0.65	0.41	0.82
Log-prob	Youden	0.22	0.86	0.35	0.62	0.25	0.78	0.38	0.72
	F1	0.45	0.48	0.47	0.93	0.10	0.04	0.05	0.96
	Recall	0.25	0.79	0.38	0.71	0.25	0.78	0.38	0.72
MAM comb. [last]	Youden	0.51	0.70	0.59	0.91	0.32	0.46	0.38	0.88
	F1	0.65	0.50	0.57	0.97	0.35	0.26	0.30	0.94
	Recall	0.20	0.95	0.33	0.51	0.20	0.74	0.32	0.64
COMET (ref.)	Youden	0.64	0.75	0.69	0.95	0.34	1.00	0.51	0.77
	F1	0.73	0.63	0.67	0.97	0.68	0.46	0.55	0.97
	Recall	0.65	0.71	0.68	0.95	0.45	0.85	0.59	0.87

Three observations follow.

1. **Precision is the bottleneck.** On Mizo, the F1-tuned LR flags 30 of 56 hallucinations with only 11 false alarms out of 444 negatives (precision 0.73, specificity

0.98). On Bodo no reference-free detector exceeds precision 0.35: to catch three quarters of the hallucinations, LR also flags 22% of non-hallucinations.

2. **Validation-chosen thresholds can fail.** With only 13–14 validation hallucinations, a threshold can land in an unrepresentative place. The clearest case is the F1 threshold for Bodo log-probability, which became so strict that it caught 2 of 54 test hallucinations. The "recall ≥ 0.8" rule also did not keep its promise on the test set for several detectors (Bodo LR: 0.65).

3. **The best rule depends on the use.** If flagged outputs are simply withheld, a high-specificity point (F1 rule) loses little. If they are sent for repair or human review, a high-recall point (Youden or recall rule) catches more hallucinations at the cost of more work.

5.4 Analysis by Hallucination Type and False-Alarm Source

Table 5.4 breaks recall down by hallucination type at the Youden threshold and adds the false-alarm rate on correct and erroneous translations separately. Figure 5.4 shows the same breakdown for Bodo with threshold-free subtype AUROC.

TABLE 5.4: Recall per hallucination type (caught / total) and false-alarm rate per negative class, at the Youden threshold. For Mizo, the type breakdown of LR and log-probability was computed only on the first extraction pass and is omitted.

		Recall by type			False-alarm rate	
Lang.	Detector	Oscillatory	Strongly det.	Fully det.	on correct	on error
Mizo	MAM combined [last]	11/16	3/7	25/33	14/243 (6%)	24/201 (12%)
	MAM FFN [last]	10/16	3/7	23/33	20/243 (8%)	23/201 (11%)
	COMET (ref.)	12/16	5/7	25/33	1/243 (0%)	23/201 (11%)
Bodo	LR	–	39/52	1/2	12/142 (8%)	85/304 (28%)
	Log-prob	–	41/52	1/2	15/142 (11%)	110/304 (36%)
	MAM combined [last]	–	24/52	1/2	6/142 (4%)	47/304 (15%)
	MAM FFN [last]	–	29/52	1/2	20/142 (14%)	80/304 (26%)
	COMET (ref.)	–	52/52	2/2	3/142 (2%)	101/304 (33%)

False alarms come from erroneous translations. In both languages, and for every detector including COMET, the false-alarm rate on *error* translations is higher than on *correct* ones, in most cases by a factor of two to four. On Bodo, log-probability flags 36% of erroneous translations but 11% of correct ones. Separating hallucinations from errors alone is also harder than the overall AUROC suggests: on Bodo the AUROC of hallucinations against errors only is 0.77 for LR and 0.75 for log-probability, against

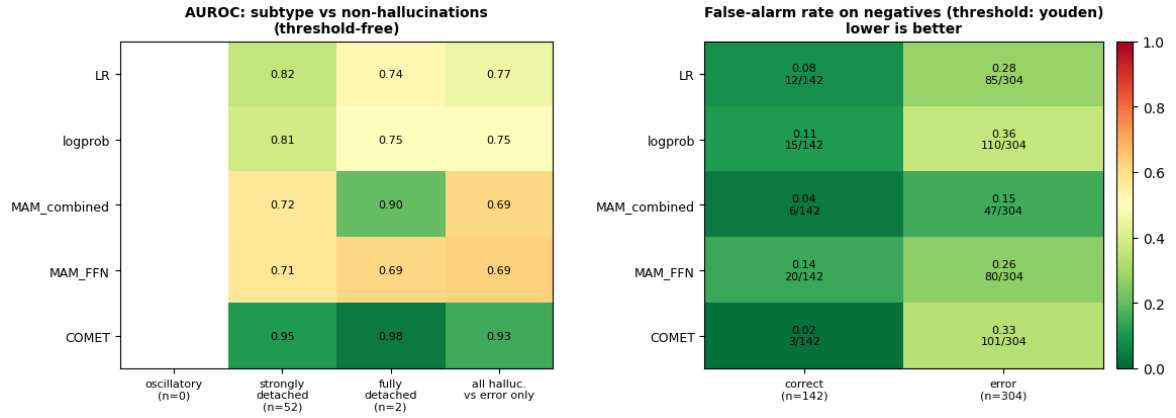


FIGURE 5.4: Bodo: AUROC of each detector for each hallucination type against all non-hallucinations (left; the last column compares hallucinations with *error* translations only) and false-alarm rate on each negative class at the Youden threshold (right).

0.82 and 0.81 when correct translations are included. In other words, the detectors largely measure *how bad* a translation is. Hallucinations are the extreme end of that scale, but ordinary errors fill the middle. For RQ2, this is the main weakness of all reference-free detectors studied here.

Types differ in difficulty. On Mizo, strongly detached hallucinations are the hardest type for the probes (3 of 7 caught), and fully detached ones the easiest. On Bodo, the MAM combined probe ranks the two fully detached hallucinations well (subtype AUROC 0.90) but strongly detached ones poorly (0.72), whereas LR and log-probability do better on strongly detached ones (0.82, 0.81). With 2–7 examples in some cells, these per-type estimates have wide confidence intervals and should be read as indicative.

Detectors complement each other. At the Youden threshold no test hallucination is missed by all five of LR, log-probability, the two MAM probes and COMET (for Mizo, using the first-pass LR and log-probability scores), while only 25 of 56 Mizo and 18 of 54 Bodo hallucinations are caught by all five.

5.5 Error Analysis

Inspecting the most confident false negatives and false positives shows recurring patterns.

Default outputs for unknown short inputs (Mizo). For several single-word Mizo entries the model outputs a generic phrase such as "in the middle" (*hmùiah*, reference "in the lips"; *dáiah*, reference "in the dew"). These are fully detached, yet COMET gives them about 0.6, because short outputs that share a frame with the reference score moderately; they are among COMET's most confident misses. Because the same default phrase recurs across unrelated inputs, such outputs are a natural target for a

simple frequency-based check.

Likely reference misalignment (Mizo). Several of the most confident misses of the reference-free LR detector are long religious passages labelled *fully detached*, where the output reads as a plausible translation of the source but the reference is unrelated. For example, a source beginning "(Sâm 49:6-9; Hebrai 2:9) ... Adama ... " is translated as "(Psalm 49:6-9; Hebrews 2:9) He offered to God the full value of the perfect life that Adam lost as a ransom", while the reference reads "Upon the return of the spies, Joshua and the people are ready to make their move". Here the label appears to reflect a misaligned source–reference pair rather than a model hallucination. A reference-based metric scores such pairs as hallucinations, while a reference-free detector correctly finds nothing wrong with the model's behaviour. Cases like this inflate the advantage of COMET in Table 5.1 and should be re-checked in the annotation.

Fluent, on-topic fabrication (Bodo). The hardest Bodo hallucinations keep the sentence frame and some source words but change the content: "The lateral line system in fish is a sensory organ that detects water vibrations" becomes "The lateral line system in the eye is a congenital organ that detects blood flow and movement". The model is confident in such outputs, so probability-based features miss them.

Garbled names (Bodo false positives). Many of LR's most confident false alarms are erroneous news headlines in which named entities are mistranslated ("Telangana's active and defensive setup kept West Bengal 39 minutes away" becomes "... kept Sonabank in a state of shock for 39 minutes"). Rare names produce low-probability tokens, which look suspicious to the detector even when the rest of the sentence is faithful.

5.6 Summary

For RQ1: reference-free detectors reach AUROC 0.896 on Mizo and 0.818 on Bodo, against 0.958 and 0.948 for COMET with a reference. Log-probability is a robust baseline in both languages; activation probes match LR on Mizo but not on Bodo, where hallucinations are fluent and training positives are few. For RQ2: detectors catch degenerate and fully detached hallucinations more reliably than strongly detached ones, and most of their false alarms are ordinary translation errors rather than correct translations.

Chapter 6

Results: Hallucination Suppression

6.1 Overview

This chapter answers RQ3: does the repair-or-abstain policy of Section 4.7 reduce the harm of hallucinations without a reference? We evaluate it with the two detectors that can score new candidates directly, log-probability and MAM combined [last], at all three validation-chosen thresholds. As *headline* setting we use the F1 threshold for Mizo and the Youden threshold for Bodo, since these give the best balance of precision and recall for the respective detectors (Table 5.3). Translation quality is measured with chrF2 against the reference (Section 4.7).

6.2 Candidate Pool

Table 6.1 summarises the alternatives generated for the 500 test sentences of each language. Every sentence received at least two distinct alternatives (on average about six), roughly half from beam search and half from sampling.

TABLE 6.1: Candidate pool for suppression (test split, 500 sentences per language).

	Mizo	Bodo
Alternatives from beam search (n-best 4, width 8)	1,670	1,542
Alternatives from nucleus sampling ($p = 0.9$, $T = 0.7$)	1,574	1,475
Total candidates incl. originals	3,744	3,517
Distinct alternatives per sentence: mean (min)	6.49 (2)	6.03 (3)
Score parity of originals: max $\lvert\Delta\rvert$ (log-prob / MAM)	0.00000 / 0.00000	0.00000 / 0.00000

6.3 What Happens to Flagged Sentences

Table 6.2 shows the outcome of the policy for every detector and threshold, split by the gold label of the original output. Figure 6.1 visualises the outcome for true hallucinations.

TABLE 6.2: Policy outcomes on the test split. H = gold hallucination (56 Mizo, 54 Bodo); non-H = correct or error (444 Mizo, 446 Bodo). "Missed" = not flagged, output kept. Headline settings in bold.

			True hallucinations			Non-hallucinations		
Lang.	Detector	Rule	Repaired	Abstained	Missed	Repaired	Abstained	Kept
Mizo	Log-prob	**F1**	8	19	29	20	13	411
		Youden	3	45	8	55	113	276
		Recall	3	41	12	47	84	313
	MAM	**F1**	10	18	28	9	6	429
		Youden	15	24	17	25	13	406
		Recall	6	47	3	87	130	227
Bodo	Log-prob	F1	2	0	52	12	7	427
		Youden	1	41	12	22	103	321
		Recall	1	41	12	22	103	321
	MAM	F1	10	4	40	21	5	420
		Youden	15	10	29	37	16	393
		Recall	11	29	14	60	100	286

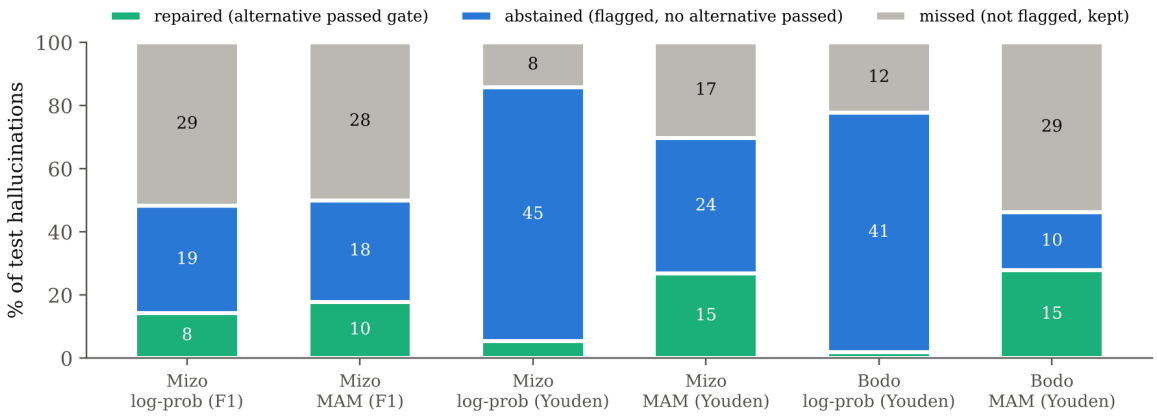


FIGURE 6.1: Fate of the true test hallucinations under the repair-or-abstain policy (numbers are sentence counts).

Two patterns are clear.

- **The log-probability gate rarely finds a passing alternative for a hallucination.** At the Youden threshold it repairs 3 of 48 flagged Mizo hallucinations and 1 of 42 flagged Bodo hallucinations; the rest end as abstentions. When the model

is unsure about a source sentence, all its candidates tend to be low-probability, so none passes the gate.

- **The mam gate accepts alternatives more readily**: 15 of 39 flagged Mizo hallucinations and 15 of 25 flagged Bodo hallucinations are repaired at the Youden threshold. Whether those repairs are actually better is the question of the next section.

Every setting also changes some non-hallucinated outputs. At the Mizo Youden threshold, the log-probability gate rewrites 55 non-hallucinated translations and withholds 113, against 3 repairs and 45 abstentions for actual hallucinations.

6.4 Effect on Translation Quality

Table 6.3 reports the mean chrF2 change produced by the policy and the three baselines on the same flagged sentences, at the headline thresholds.

TABLE 6.3: Mean change in chrF2 (final − original) with 95% bootstrap CIs, headline thresholds (Mizo: F1; Bodo: Youden). "All" averages over all 500 test sentences (unchanged sentences count as 0). Flag./chg. = sentences flagged / changed.

Lang.	Det.	Method	Flag./chg.	All (500)	True H	Non-H
Mizo	LP	policy	60/28	+0.01 [−0.21, +0.23]	−0.18 [−0.77, +0.25]	+0.04 [−0.18, +0.25]
		forced_best	60/60	−0.29 [−0.65, +0.05]	−1.11 [−2.29, −0.05]	−0.18 [−0.57, +0.16]
		random	60/60	−0.28 [−0.64, +0.04]	−0.52 [−1.22, +0.09]	−0.25 [−0.67, +0.11]
		oracle	60/47	+0.54 [+0.36, +0.78]	+1.72 [+0.92, +2.64]	+0.39 [+0.21, +0.63]
	MAM	policy	43/19	+0.02 [−0.23, +0.31]	−0.03 [−1.13, +0.98]	+0.03 [−0.22, +0.31]
		forced_best	43/43	−0.10 [−0.38, +0.19]	−0.58 [−1.98, +0.72]	−0.04 [−0.31, +0.26]
		random	43/43	−0.01 [−0.31, +0.37]	−0.62 [−1.72, +0.40]	+0.07 [−0.22, +0.48]
		oracle	43/34	+0.56 [+0.28, +0.94]	+1.93 [+1.06, +2.92]	+0.39 [+0.09, +0.82]
Bodo	LP	policy	167/23	−0.05 [−0.20, +0.09]	−0.03 [−0.08, +0.00]	−0.05 [−0.22, +0.10]
		forced_best	167/167	−0.21 [−0.54, +0.13]	−1.32 [−2.48, −0.20]	−0.07 [−0.41, +0.26]
		random	167/167	−0.49 [−0.95, −0.02]	−0.67 [−1.92, +0.63]	−0.47 [−0.98, +0.00]
		oracle	167/139	+2.09 [+1.68, +2.52]	+4.94 [+3.30, +6.82]	+1.74 [+1.32, +2.17]
	MAM	policy	78/52	−0.45 [−0.82, −0.09]	−0.34 [−1.46, +0.81]	−0.46 [−0.91, −0.11]
		forced_best	78/78	−0.63 [−1.07, −0.25]	−0.60 [−2.04, +0.69]	−0.64 [−1.11, −0.26]
		random	78/78	−0.34 [−0.65, −0.04]	−0.81 [−2.13, +0.40]	−0.28 [−0.64, +0.01]
		oracle	78/66	+0.96 [+0.68, +1.25]	+2.82 [+1.54, +4.22]	+0.74 [+0.48, +1.02]

Repairs do not improve quality on average. The gated policy changes mean chrF2 by between −0.45 and +0.02 points, and three of the four confidence intervals include zero. The one clearly negative result is the MAM gate on Bodo (−0.45), whose repairs make translations worse. Repairs of true hallucinations are no better: none of the four policy rows shows a significant gain for the true-hallucination subset.

Gating is better than not gating. Replacing every flagged output with the detector's favourite alternative (*forced_best*) is worse than the gated policy in all four cases, and significantly harmful for true hallucinations under the log-probability detector (Mizo −1.11, Bodo −1.32). The gate's refusal to accept a still-suspicious alternative is therefore doing useful work.

Better translations exist in the pool, but the detectors cannot find them. The oracle, which picks the alternative closest to the reference, would add 0.5–2.1 chrF2 overall and 1.7–4.9 chrF2 on true hallucinations. The candidate pool contains real improvements; the bottleneck is selecting them. A detector score measures how *typical* an output looks to the model, which is not the same as how *faithful* it is to the source.

Sentence-level verdicts for Bodo at the Youden threshold make this concrete (a change of more than 1 chrF2 point counts as better or worse). Of the 23 log-probability repairs, 12 were better, 1 unchanged and 10 worse. Of the 52 MAM repairs, only 12 were better, 8 unchanged and 32 worse; among the 15 MAM repairs of true hallucinations, 3 were better, 4 unchanged and 8 worse.

6.5 Do the Two Detectors Agree on Repairs?

If a repair is genuinely better, an independent detector should also accept it. Table 6.4 checks how many repairs chosen by one detector are still flagged by the other.

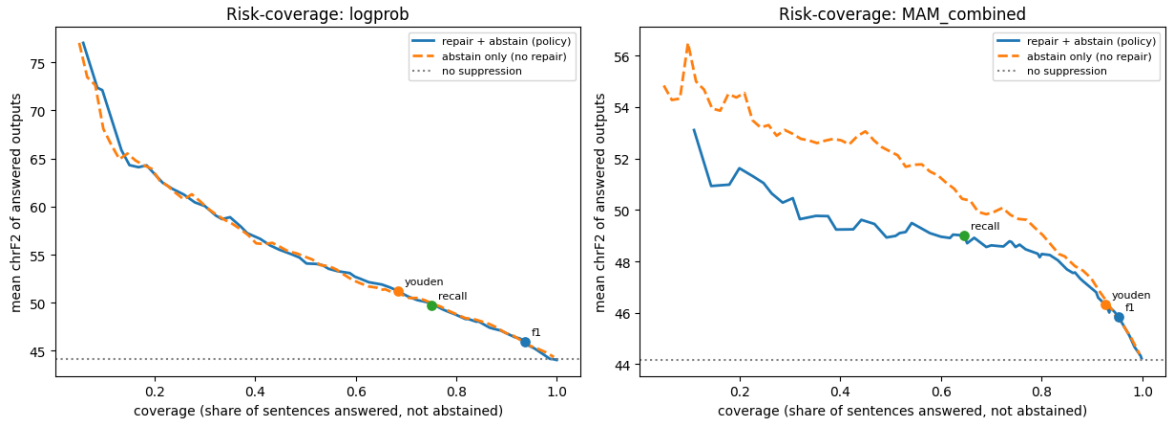
TABLE 6.4: Repairs selected by one detector that the other detector still flags (headline thresholds).

	Mizo (F1)	Bodo (Youden)
Selected by log-prob, still flagged by MAM	8 of 28 (28.6%)	4 of 23 (17.4%)
Selected by MAM, still flagged by log-prob	14 of 19 (73.7%)	39 of 52 (75.0%)

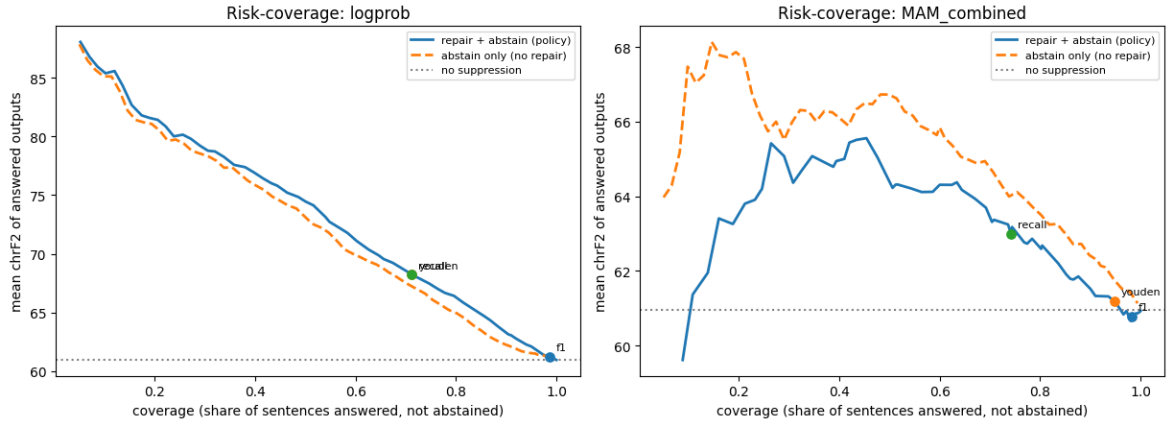
Repairs chosen by the log-probability gate mostly satisfy MAM too. The reverse is not true: about three quarters of MAM-selected repairs remain suspicious to the log-probability detector. Together with the quality numbers above, this suggests that the MAM probe can be satisfied by alternatives that change the decoder state without making the translation more faithful, for example by switching to a different fabricated content (Section 6.8). For the Mizo log-probability repairs, 20 of 28 also pass MAM, and 3 are noticeably more repetitive than the original (trigram repetition up by more than 0.05), a known way to raise per-token probability.

6.6 Risk–Coverage: Abstention Is the Useful Part

Figure 6.2 plots the mean chrF2 of the outputs the system still answers against the fraction of sentences it answers, as the threshold varies. The dotted line is the quality without suppression (about 44 chrF2 for Mizo and 61 for Bodo). The solid line is the full policy; the dashed line abstains on every flagged sentence without attempting repair.



(a) Mizo



(b) Bodo

FIGURE 6.2: Risk–coverage curves: mean chrF2 of answered outputs against coverage, for the full policy (solid), abstention only (dashed) and no suppression (dotted). Dots mark the three validation-chosen thresholds.

Both detectors produce useful rankings: as coverage drops, the quality of the remaining outputs rises steadily. At the Youden threshold the log-probability detector answers about 68% of Mizo and 71% of Bodo sentences, and the answered outputs score roughly 51 and 68 chrF2, about 7 points above the unfiltered baselines. For log-probability the two curves coincide, because the gate repairs very few sentences. For MAM, the abstain-only curve lies *above* the full policy at almost every coverage level in both languages: the system would be better off withholding flagged outputs than replacing

them with the alternatives the probe accepts.

6.7 End-to-End Inference

The offline evaluation was confirmed with a stand-alone, reference-free inference function that takes only a source sentence, translates it, scores it, and repairs or abstains. Run on the 500 Mizo test sentences with their stored outputs as originals (MAM, F1 threshold), it reproduced the offline result exactly: 457 kept (28 of them hallucinations), 19 repaired (10 hallucinations) and 24 abstained (18 hallucinations). On Bodo, with fresh decoding and fresh sampling, it gave 335 kept, 22 repaired and 143 abstained for log-probability, and 422 kept, 54 repaired and 24 abstained for MAM (offline: 333/23/144 and 422/52/26); the small differences come from sampling. Processing all 500 Bodo sentences with both detectors took 16.8 minutes on a single T4 GPU, about one second per sentence per detector.

6.8 Examples

Table 6.5 illustrates the behaviours discussed above. Good repairs exist, both for real hallucinations and for small lexical errors. But the gate can also swap one fabrication for another ("snowflakes" becomes "mountain gorillas") or accept a degenerate repetitive output, because repetition makes each token highly predictable and raises the length-normalised log-probability.

6.9 Summary

For RQ3: a reference-free gate does not make repairs reliably better. Averaged over the test set, chrF2 changes by between -0.45 and $+0.02$, while an oracle choosing from the same candidates would gain 0.5–2.1. The gate is still clearly better than replacing every flagged output, and the detectors rank outputs well enough that *abstaining* on flagged sentences raises the quality of what is shown by about 7 chrF2 points at around 70% coverage. In practice, the safest use of these detectors is to flag or withhold unreliable translations, not to rewrite them automatically.

TABLE 6.5: Examples of policy decisions. Δ = chrF2 change (analysis only; the reference is not used by the policy).

Successful repair (Bodo R0976, strongly detached; MAM gate, also passes log-prob; $\Delta = +15.4$)	
Reference	The lateral line system in fish is a sensory organ that detects water vibrations and movement.
Original	The lateral line system in the eye is a congenital organ that detects blood flow and movement.
Repair	The lateral line system in the eye is a sensory organ that detects water movement and movement.
Repair of a near-miss (Bodo R2965, correct; log-prob gate, passes MAM; $\Delta = +7.4$)	
Reference	Assam to Introduce Smart Agriculture Solutions for Local Farmers
Original	Assam to Introduce Smart Agrical Solutions for Local Farmers
Repair	Assam to Introduce Smart Agricultural Solutions for Local Farmers
One hallucination swapped for another (Bodo; MAM gate)	
Original	Although snowflakes die for a variety of reasons, the death of two-quarters of them is caused by severe weather.
Repair	Although mountain gorillas die for many reasons, half of the deaths are caused by severe environmental conditions.
Repair that introduces oscillation (Mizo R0893, strongly detached; log-prob gate; $\Delta = -13.3$)	
Reference	The mortal moon hath her eclipse endured,
Original	When the dead moon lifts up the image of the dead,
Repair	When the dead moon is alive, when the dead moon is alive, when the dead moon is alive, ... (the phrase keeps repeating)
Abstention (Bodo R0529; MAM gate; all 7 alternatives still flagged)	
Original	James Princip, a maker of textiles from the East India Company, invented the Brahmi and Charoesti, two tools used in early pottery and pottery.

Chapter 7

Results: Round-Trip Features for Bodo

7.1 Overview

The detectors of Chapter 5 ask the translation model to judge its own output. A model that is confidently wrong will also be confidently wrong about its confidence, which is one reason fluent Bodo hallucinations are hard to catch. This chapter answers RQ4 by adding an *independent* judge: a separately trained English→Bodo model (Section 4.8) that measures how well the English output explains the original Bodo source.

The experiments use their own label-stratified split of the Bodo data: 1,191 training, 298 validation and 500 test sentences, with 54 test hallucinations (51 strongly and 3 fully detached). The base internal-feature LR is retrained on this split, so its AUROC (0.823) differs slightly from Chapter 5 (0.818).

7.2 Do Round-Trip Features Separate Hallucinations?

Averaged over all 1,989 sentences, the round-trip signals behave as expected. The mean per-token log-probability of the Bodo source given the English output is -2.40 for hallucinations against -1.50 for other outputs; the PMI is 2.37 against 3.16 (a hallucinated output raises the probability of the source much less above its prior); and the chrF of the back-translation against the source is 41.1 against 54.1.

Table 7.1 and Figure 7.1 give the test results.

TABLE 7.1: Round-trip detectors on the Bodo round-trip test split (500 sentences, 54 hallucinations). AUROC with 95% bootstrap CI; P, R, F1 at the validation-chosen Youden threshold; P@R80 = best precision at recall ≥ 0.8.

Detector	AUROC	95% CI	P	R	F1	P@R80
single: `rt_lp_mean`	0.860	[0.816, 0.904]	0.23	0.91	0.37	0.29
single: `rt_pmi_mean`	0.757	[0.693, 0.828]	0.23	0.70	0.34	0.19
single: `rt_lp_min`	0.680	[0.611, 0.744]	0.14	0.91	0.25	0.15
single: `rt_chrf`	0.791	[0.730, 0.847]	0.19	0.87	0.32	0.22
single: `rt_bleu`	0.717	[0.635, 0.791]	0.13	0.85	0.23	0.17
LR: round-trip only (7 features)	0.861	[0.800, 0.906]	0.19	0.94	0.32	0.28
LR: base internal (19 features)	0.823	[0.765, 0.875]	0.28	0.76	0.40	0.28
LR: base + round-trip (26 features)	**0.865**	[0.816, 0.906]	0.28	0.87	**0.42**	**0.29**

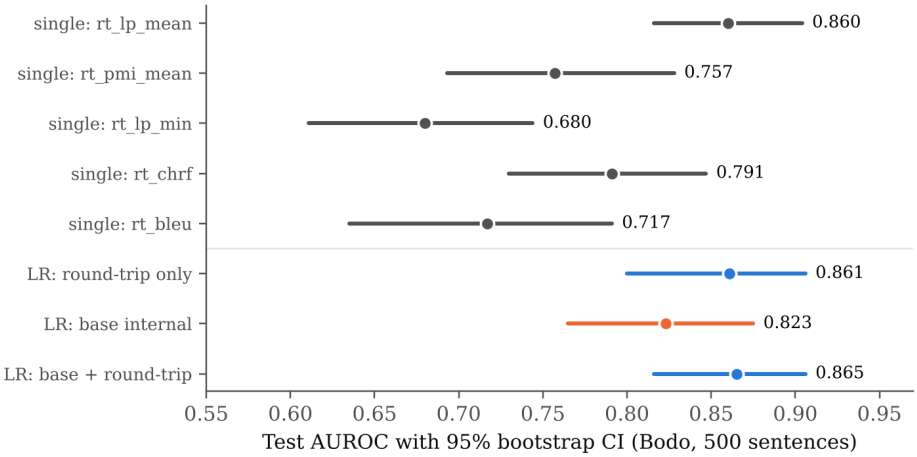


FIGURE 7.1: Test AUROC with 95% bootstrap confidence intervals for the round-trip detectors (Bodo).

The single best round-trip feature, `rt_lp_mean`, reaches AUROC 0.860 on its own. That is higher than the 19-feature internal LR (0.823) and higher than any single internal feature in Chapter 5 (best: `seq_logprob`, 0.805). Forced decoding of the source works better than free back-translation (chrF 0.791, BLEU 0.717): scoring the actual source under the reverse model avoids the noise of a second decoding step. The PMI correction does not help on average, although it ranks the three fully detached hallucinations very well (subtype AUROC 0.964).

7.3 Added Value over Internal Features

Combining the round-trip features with the 19 internal features gives the best Bodo detector in this project, with AUROC 0.865. A paired bootstrap over the test set gives an AUROC gain of +0.042 over the base LR, with a 95% interval of $[-0.010, +0.096]$; the gain is positive in 95% of resamples. The improvement is consistent but, with 54

test hallucinations, not conclusive at the 5% level. Precision at fixed recall improves at every level examined (Table 7.2).

TABLE 7.2: Precision at fixed recall, base internal LR versus base + round-trip LR (Bodo).

	Recall ≥ 0.7	Recall ≥ 0.8	Recall ≥ 0.9
Base internal	0.30	0.28	0.22
Base + round-trip	0.35	0.29	0.27

Figure 7.2 shows the confusion matrices at the Youden threshold. Adding round-trip features raises recall from 41 to 47 of 54 hallucinations (strongly detached: 39 to 44 of 51; fully detached: 2 to 3 of 3) at the cost of 14 additional false alarms (108 to 122). The round-trip-only model catches 51 of 54 hallucinations but flags almost half of all non-hallucinations. As in Chapter 5, the false alarms fall mainly on erroneous translations: the combined model flags 13 of 143 correct translations but 109 of 303 erroneous ones.

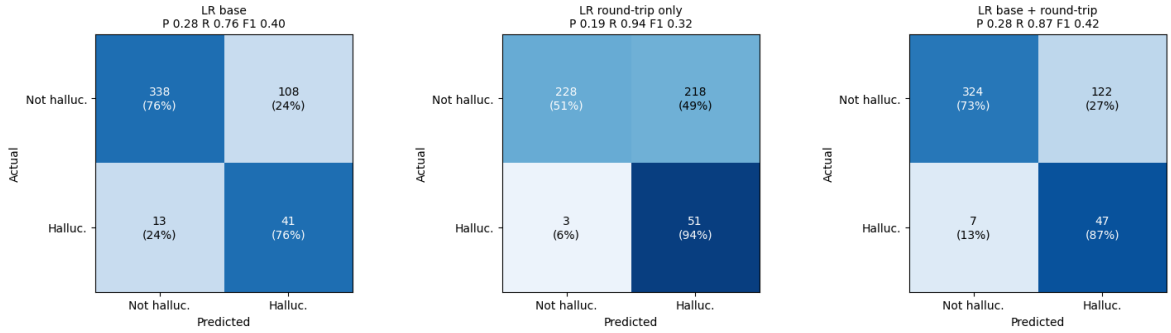


FIGURE 7.2: Test confusion matrices at the validation-chosen Youden threshold for the base internal LR, the round-trip-only LR and the combined LR (Bodo).

7.4 Summary

For RQ4: an independent reverse model adds real signal. Its forced-decoding probability of the source is the strongest single reference-free feature we found for Bodo, and adding round-trip features raises AUROC from 0.823 to 0.865, recall at the Youden threshold from 76% to 87%, and precision at high recall. The gain over the internal-feature detector is positive in 95% of bootstrap resamples, but its confidence interval still includes zero. The cost is a second model that must be trained and run for every sentence.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This project studied how to detect and handle hallucinations in Mizo→English and Bodo→English translation with fine-tuned NLLB-200 models, using only signals available at inference time. The answers to the four research questions are as follows.

RQ1 (detection). Reference-free detectors are useful but clearly weaker than a reference-based metric. On Mizo, the activation probe (MAM combined [last], AUROC 0.896) and logistic regression over 19 internal features (0.894) perform on a par with chrF2 computed against the reference. On Bodo, logistic regression reaches 0.818, with mean token log-probability alone at 0.805. COMET, which needs a reference, reaches about 0.95 in both languages. The token log-probability of the output is the most transferable single signal; attention, logit-lens and repetition features help in one language but not the other.

RQ2 (what is caught and missed). Degenerate outputs (repetition, generic defaults for unknown short inputs) and fully detached hallucinations are caught most reliably. Fluent, on-topic strongly detached hallucinations, which make up almost all Bodo hallucinations, are the hardest case. Most false alarms are ordinary translation errors rather than correct translations, so the detectors largely measure how bad a translation is rather than whether it is fabricated. With 6–8% of outputs hallucinated but 43–64% erroneous, this limits achievable precision.

RQ3 (suppression). Re-decoding flagged sentences and accepting an alternative that passes the detector does not reliably improve translations (chrF2 change between -0.45 and $+0.02$), even though an oracle choosing from the same candidates would gain 0.5–2.1 chrF2. Gating is still much better than replacing every flagged output.

The practical value lies in abstention: withholding flagged outputs raises the quality of what is shown by about 7 chrF2 points at roughly 70% coverage.

RQ4 (round-trip). An independent English→Bodo model provides the strongest single reference-free feature found for Bodo (forced-decoding log-probability of the source, AUROC 0.860). Adding round-trip features to the internal ones raises AUROC from 0.823 to 0.865 and recall at the Youden threshold from 76% to 87%; the gain is positive in 95% of bootstrap resamples but not yet significant at the 5% level.

Overall, a deployable system for these languages should use the detectors to *flag* unreliable translations—for a warning, human review, or withholding—rather than to rewrite them silently.

8.2 Limitations

- **Small test sets.** Each test set has 54–56 hallucinations, and some type-level cells have 2–7. Confidence intervals are wide, and type-level results are indicative only.
- **Single split.** All results use one fixed split (seed 42). The code supports repeated random splits, but they were not run; variance across splits is therefore not reported.
- **Few validation positives.** Thresholds are chosen on 13–14 validation hallucinations, which produced at least one degenerate threshold (Bodo log-probability, F1 rule).
- **Label quality.** Some Mizo items labelled as fully detached appear to have misaligned references rather than hallucinated outputs. No inter-annotator agreement is reported.
- **Quality metric for repairs.** Repairs were judged with chrF2 because the COMET step failed in both suppression runs. chrF2 rewards surface overlap and may undervalue valid paraphrases.
- **Mixed extraction passes for Mizo.** The Mizo subtype analysis for LR and log-probability and the Mizo ROC figure come from the first feature-extraction pass; all other Mizo results come from the re-extraction described in Section 4.7.4.
- **Mizo decoding mismatch.** Re-decoding reproduces only 50% of the stored Mizo outputs with 4-beam search, so the end-to-end pipeline's "original" translation does not always match the labelled one.
- **Round-trip only for Bodo.** The reverse-model experiment was run for Bodo only, on its own split.

8.3 Future Work

- **Repeated splits and calibration.** Re-run all detectors over several random splits and report mean and spread; choose thresholds by cross-validation on the fit and validation data together, to avoid degenerate thresholds.
- **Separating hallucinations from errors.** Train detectors with a three-way objective (correct / error / hallucination), or add features that target fabricated content specifically, such as source-contribution measures (Dale et al., 2023a) and named-entity consistency between source and output.
- **Better repair selection.** Use the round-trip score, or a combination of detectors, to choose among candidates; the oracle results show there is headroom of 0.5–2.1 chrF2. Penalise repetition explicitly so that the gate cannot be satisfied by degenerate outputs.
- **Round-trip for Mizo** and joint training of a single detector over both languages to test transfer.
- **Annotation.** Re-check items with likely reference misalignment, collect a second annotation for agreement statistics, and enlarge the hallucination set (particularly strongly detached Mizo and oscillatory Bodo examples).
- **COMET-based evaluation** of repairs, with candidate scoring batched to fit in GPU memory, and a small human evaluation of repaired outputs.

8.4 Closing Remarks

For languages like Mizo and Bodo, every translation a reader cannot verify carries risk. This work shows that a translation model's own internal signals, supplemented by an independent reverse model, can identify a large share of hallucinated outputs without any reference, and that the most dependable use of that ability is to tell the reader when not to trust a translation.

Appendix A

Feature Definitions and Experimental Configuration

A.1 Formal Feature Definitions

Let $x = (x_1, \ldots, x_S)$ be the source tokens and $y = (y_1, \ldots, y_T)$ the output tokens after the forced English language token. Let $p_\theta(y_i \mid y_{<i}, x)$ be the model's probability of the i-th output token in teacher-forced decoding, $A \in \mathbb{R}^{T \times S}$ the final-layer cross-attention averaged over heads (each row renormalised to sum to 1), and $h_i^{(\ell)}$ the decoder hidden state at layer ℓ and position i.

$$
\begin{aligned}
\texttt{seq_logprob} &= \frac{1}{T}\sum_{i=1}^{T} \log p_\theta(y_i \mid y_{<i}, x) \\
\texttt{min_token_logprob} &= \min_i \log p_\theta(y_i \mid y_{<i}, x) \\
\texttt{source_entropy} &= \frac{1}{T}\sum_{i=1}^{T} \frac{-\sum_j A_{ij} \log A_{ij}}{\log S} \\
\texttt{source_concentration} &= \frac{1}{T}\sum_{i=1}^{T} \max_j A_{ij} \\
\texttt{top20_mass} &= \frac{1}{T}\sum_{i=1}^{T} \sum_{j \in \text{top-}k(A_{i\cdot})} A_{ij}, \quad k = \lceil 0.2\, S \rceil \\
\texttt{high_contrib_ratio} &= \frac{1}{S}\sum_{j=1}^{S} \mathbb{1}\Big[\textstyle\sum_i A_{ij} \geq 0.5\Big] \\
\texttt{staticity_w}_w &= \frac{1}{T-w}\sum_{i=1}^{T-w} \cos(A_{i\cdot}, A_{i+w,\cdot}), \quad w \in \{1,2,3\} \\
\texttt{anchor_layer_entropy}_\ell &= \frac{1}{T}\sum_{i=1}^{T} H\Big(\text{softmax}(W_{\text{out}}\,\text{LN}(h_i^{(\ell)}))\Big), \quad \ell \in \{6,9,12\} \\
\texttt{subword_drift_mean} &= \frac{1}{T-1}\sum_{i=1}^{T-1} \Big(1 - \cos(h_i^{(L)}, h_{i+1}^{(L)})\Big) \\
\texttt{ngram_repetition_n} &= 1 - \frac{\#\text{unique word } n\text{-grams}}{\#\text{word } n\text{-grams}} \\
\texttt{length_ratio} &= T/S
\end{aligned}
$$

Here $H(\cdot)$ is Shannon entropy, LN the decoder's final layer normalisation (applied to intermediate layers before projection, as in the logit lens), and W_{out} the output projection. `anchor_layer_entropy_mean` and `_min` are the mean and minimum over the three anchor layers; `staticity_max` is the maximum over w.

A.2 Experimental Configuration

TABLE A.1: Experimental configuration.

Setting	Value
Translation models	NLLB-200-distilled-600M fine-tuned for Mizo→English and romanised Bodo→English (12 decoder layers, $d_{\text{model}} = 1024$)
Language codes	source `lus_Latn` (Mizo; also used for Bodo), target `eng_Latn`
Max. sequence length	128 tokens (feature extraction, MAM, suppression)
Text cleaning	`ftfy` repair, non-breaking spaces removed, whitespace collapsed
Split	seed 42; test 500 rows with 45% of hallucinations; validation 20% of rest with 20% of remaining hallucinations
Single-feature direction	sign chosen on fit split (AUROC ≥ 0.5)
LR	median imputation, standardisation, L_2, $C = 1$, balanced class weights, max. 5,000 iterations
MAM probe	final decoder layer; SA / CA / FFN outputs at first and last position; MLP 256 ReLU; Adam, lr 10^{-4}, weight decay 10^{-2}; batch 128; 1,000 iterations; BCE with positive weight = neg/pos
Thresholds	Youden; max-F1; max precision s.t. recall ≥ 0.8 — all on validation
Candidates	beam width 8, n-best 4; nucleus sampling $p = 0.9$, $T = 0.7$, 4 samples
Suppression detectors	$-$`seq_logprob`; MAM combined [last]
Quality metric	chrF2 (sacreBLEU); 2,000 bootstrap resamples
Reverse model	NLLB-200-distilled-600M, English→Bodo, target code `asm_Beng`; 3 epochs; batch 8×4 accumulation; lr 5×10^{-5}, linear decay, 20 warm-up steps; Adafactor; fp16
Back-translation	4-beam, max. 128 new tokens
Round-trip split	stratified by label; test 500, validation 298, train 1,191
Reference metrics	sentence BLEU (effective order), chrF2, COMET `Unbabel/wmt22-comet-da`
Hardware	Kaggle, NVIDIA T4 GPU (16 GB)

Bibliography

Azaria, A. and Mitchell, T. (2023). The internal state of an LLM knows when it's lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976.

Dale, D., Voita, E., Barrault, L., and Costa-jussà, M. R. (2023a). Detecting and mitigating hallucinations in machine translation: Model internal workings alone do well, sentence similarity even better. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 36–50.

Dale, D., Voita, E., Lam, J., Hansanti, P., Ropers, C., Kalbassi, E., Gao, C., Barrault, L., and Costa-jussà, M. R. (2023b). HalOmi: A manually annotated benchmark for multilingual hallucination and omission detection in machine translation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 638–653.

Gala, J., Chitale, P. A., AK, R., et al. (2023). IndicTrans2: Towards high-quality and accessible machine translation models for all 22 scheduled Indian languages. *Transactions on Machine Learning Research*.

Geifman, Y. and El-Yaniv, R. (2017). Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30.

Guerreiro, N. M., Alves, D. M., Waldendorf, J., Haddow, B., Birch, A., Colombo, P., and Martins, A. F. T. (2023a). Hallucinations in large multilingual translation models. *Transactions of the Association for Computational Linguistics*, 11:1500–1517.

Guerreiro, N. M., Colombo, P., Piantanida, P., and Martins, A. F. T. (2023b). Optimal transport for unsupervised hallucination detection in neural machine translation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 13766–13784.

Guerreiro, N. M., Voita, E., and Martins, A. F. T. (2023c). Looking for a needle in a haystack: A comprehensive study of hallucinations in neural machine translation. In

Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL), pages 1059–1075.

Holtzman, A., Buys, J., Du, L., Forbes, M., and Choi, Y. (2020). The curious case of neural text degeneration. In *International Conference on Learning Representations (ICLR)*.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., and Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.

Lee, K., Firat, O., Agarwal, A., Fannjiang, C., and Sussillo, D. (2019). Hallucinations in neural machine translation. In *International Conference on Learning Representations (ICLR), OpenReview submission*.

Moon, J., Cho, H., and Park, E. L. (2020). Revisiting round-trip translation for quality estimation. In *Proceedings of the 22nd Annual Conference of the European Association for Machine Translation (EAMT)*.

NLLB Team, Costa-jussà, M. R., Cross, J., et al. (2022). No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*.

nostalgebraist (2020). Interpreting GPT: The logit lens. LessWrong blog post.

Papineni, K., Roukos, S., Ward, T., and Zhu, W.-J. (2002). BLEU: A method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 311–318.

Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

Popović, M. (2015). chrF: Character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation (WMT)*, pages 392–395.

Post, M. (2018). A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation (WMT)*, pages 186–191.

Raunak, V., Menezes, A., and Junczys-Dowmunt, M. (2021). The curious case of hallucinations in neural machine translation. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, pages 1172–1183.

Rei, R., C. de Souza, J. G., Alves, D., Zerva, C., Farinha, A. C., Glushkova, T., Lavie, A., Coheur, L., and Martins, A. F. T. (2022). COMET-22: Unbabel-IST 2022 submission for the metrics shared task. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 578–585.

Rei, R., Stewart, C., Farinha, A. C., and Lavie, A. (2020). COMET: A neural framework for MT evaluation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2685–2702.

Snyder, B., Moisescu, M., and Zafar, M. B. (2024). On early detection of hallucinations in factual question answering. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.

Tu, Z., Liu, Y., Shang, L., Liu, X., and Li, H. (2017). Neural machine translation with reconstruction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 31.

Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212.

Wolf, T., Debut, L., Sanh, V., et al. (2020). Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45.

Xu, W., Agrawal, S., Briakou, E., Martindale, M. J., and Carpuat, M. (2023). Understanding and detecting hallucinations in neural machine translation via model introspection. *Transactions of the Association for Computational Linguistics*, 11:546–564.

Youden, W. J. (1950). Index for rating diagnostic tests. *Cancer*, 3(1):32–35.

Zhou, C., Neubig, G., Gu, J., Diab, M., Guzmán, F., Zettlemoyer, L., and Ghazvininejad, M. (2021). Detecting hallucinated content in conditional neural sequence generation. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 1393–1404.